

LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 4
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Date: 30-08-2026

1. TITLE

Dynamic Webpage Manipulation Using JavaScript DOM Functions and Event Listeners

2. AIM

To create an interactive webpage utilizing JavaScript DOM methods (getElementById, createElement, appendChild) and event listeners for dynamic heading changes, background color updates, and list item additions.

3. OBJECTIVE

The objectives of this experiment are:

- To select DOM elements using document.getElementById.
- To register event listeners using addEventListener.
- To modify document.body.style.backgroundColor dynamically.
- To append new li elements to an unordered list.

4. THEORY

The Document Object Model (DOM) represents web pages as node trees. JavaScript methods inspect and mutate element properties and inline styles dynamically.

Application Flow:

User Action -> Event Listener -> DOM Function Call -> Style/Element Mutation -> Immediate Page Update

5. ALGORITHM / PROCEDURE

1. Create week4.html with heading, buttons, color input, item input, and list.
2. Write changeText() to update heading innerHTML.
3. Write changeColor() to update document.body.style.backgroundColor.
4. Write addItem() to create and append li elements to #myList.
5. Attach click listeners to buttons and test functionality.

6. PROGRAM

week4.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Dynamic Webpage Using DOM & JS</title>
  <style>
    body{
      font-family: Arial;
      text-align: center;
      background-color: lightblue;
    }

    h1{
      color: darkblue;
    }

    button{
      padding: 10px 20px;
      margin: 10px;
      font-size: 16px;
    }

    ul{
      list-style-type: square;
      width: 200px;
      margin: auto;
      text-align: left;
    }
  </style>
</head>

<body>

  <h1 id="heading">Welcome to DOM Example</h1>

  <button id="btn1">Change Heading</button><br>
  <button id="btn2">Change Background</button>
  <input type="text", id="newcolor" , placeholder ="Enter color"><br>
  <button id="btn3">Add List Item</button>
  <input type="text", id="newitem" , placeholder ="Enter item">

  <ul id="myList">
    <li>IronMan</li>
    <li>Hulk</li>
  </ul>

  <script>

function changeText() {
  var text = document.getElementById("heading");
  text.innerHTML = "Dynamic Webpage Using HTML, CSS & JavaScript";
}

function changeColor() {
  var color = document.getElementById("newcolor").value;
  document.body.style.backgroundColor = color;
  document.getElementById("newcolor").value="";
}
```

```
}

function addItem() {
    var text = document.getElementById("newitem").value;
    var item = document.createElement("li");
    item.innerHTML = text;
    document.getElementById("myList").appendChild(item);
    document.getElementById("newitem").value="";
}

var changetext = document.getElementById("btn1");
changetext.addEventListener("click", changeText);
var changecolor = document.getElementById("btn2");
changecolor.addEventListener("click", changeColor);
var additem = document.getElementById("btn3");
additem.addEventListener("click", addItem);

</script>

</body>
</html>
```

7. CODE EXPLANATION

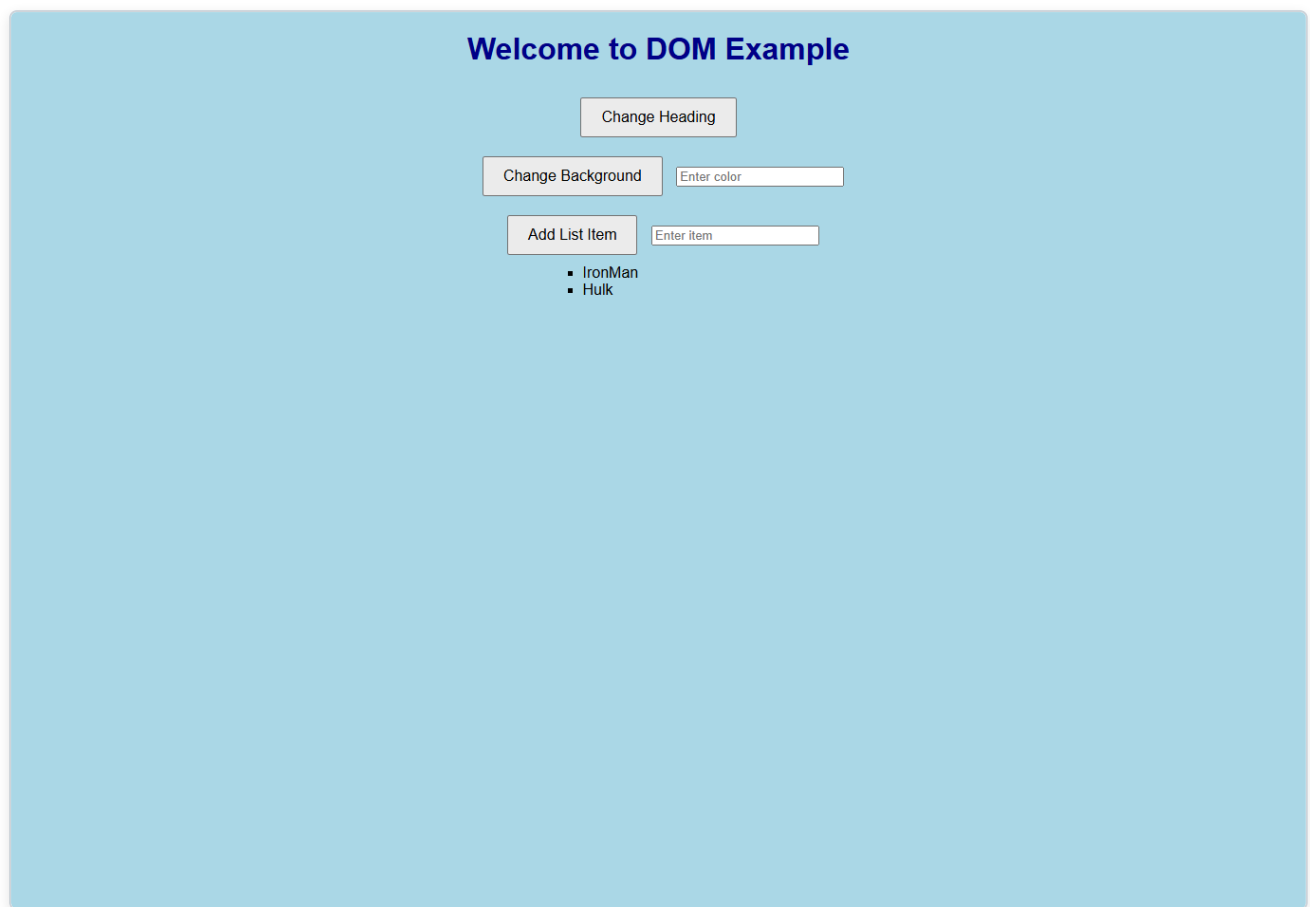
Code / Syntax	Explanation
document.getElementById("heading")	Selects target element node by its ID attribute.
document.body.style.backgroundColor	Dynamically changes document body background color.
document.createElement("li")	Creates new list item DOM node.
myList.appendChild(item)	Appends new child node to unordered list.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Change Heading	Click Change Heading button	Heading text updates dynamically	Pass
TC-02	Change Background	Enter color & click button	Body background color changes to input value	Pass
TC-03	Add List Item	Enter item & click button	New item appended to list	Pass

9. OUTPUT



Output 1: Dynamic Webpage Manipulation Using JavaScript DOM Functions and Event Listeners Rendered Webpage

10. RESULT

Thus, dynamic webpage manipulation using javascript dom functions and event listeners was successfully implemented and verified across all test cases.